

Milestone 3

1. Project Objective.

The objective of this milestone is to study and present the spatial distribution of natural disaster declarations across the United States. Interns will examine how disaster occurrences differ across states and regions, identifying areas that experience a higher frequency of such events.

This phase emphasizes transforming summarized data into geographical representations, such as maps and visual tools, to uncover patterns that are not easily visible in raw tables. As outlined in the project plan, the task includes grouping data based on location and creating visualizations like choropleth maps and state-wise comparisons to better understand disaster trends across the country.

2. Problem Formulation.

The problem formulation for this milestone is based on **Geospatial Data Analysis and Exploratory Data Analysis (EDA)** to understand the distribution of natural disaster declarations across different states in the United States.

Geospatial analysis focuses on examining location-based patterns and identifying regions with higher disaster occurrences, while EDA helps in cleaning, summarizing, and exploring the dataset. Data aggregation techniques are used to group disaster records by state for meaningful comparison.

Visualization methods such as choropleth maps are applied to represent disaster intensity using color variations. This approach helps in identifying geographical hotspots and uncovering patterns that are not easily visible in raw data.

3. Dataset Preparation for Geographic Analysis

For this milestone, the dataset used is **usnd_cleaned.csv**, which is the processed and cleaned output generated in Milestone-1. This dataset serves as the foundation for performing geospatial analysis and visualization. To ensure accurate analysis, certain key columns are required, including **state**, **incident Type**, **year**, and **month**, as they help in identifying the location, type, and time of disaster occurrences. Additionally, optional columns such as **state Name** and **declaration Date** can enhance the analysis by providing more descriptive and

time-specific information. Proper selection and preparation of these fields enable effective aggregation and mapping of disaster data across different states.

4.Data Validation.

Data validation ensures that the dataset is accurate, complete, and reliable before analysis. The code `df.is null().sum()` checks for missing values in each column, helping identify incomplete data. The statement `df ['State'].unique()` shows all unique state names to detect any inconsistencies or errors. Finally, `len(df)` gives the total number of records, ensuring the dataset size is correct. These steps help confirm that the data is clean and ready for further geospatial analysis.

5. State-Level Aggregation.

This step focuses on data aggregation to understand the distribution of disasters across different states. The code uses the `groupby()` method, which is a Pandas function used to group data based on a specific column—in this case, State. After grouping, the `size()` method is applied to count the number of disaster records for each state. The results are then sorted in descending order using `sort_values(ascending=False)` to identify the states with the highest number of disaster declarations.

The output shows that states like Texas (TX), Missouri (MO), and Kentucky (KY) have the highest number of disasters, indicating potential geographical hotspots. Finally, `head()` is used to display the top 5 states. This approach helps in comparing disaster frequency across states and supports further geospatial visualization.

6. State + Incident Type Aggregation.

This step involves data transformation and multi-level aggregation to analyze how different types of disasters are distributed across states. The column is first renamed using the `rename()` method to improve clarity, and the dataset structure is verified using `df.columns`. The `groupby()` method is then applied on both State and Incident type to group the data, while the `size()` method counts the number of occurrences for each combination. The `unstack()` method reshapes the result into a matrix format, where each disaster type becomes a separate column and missing values are filled with zero.

The output is a structured table showing the frequency of each disaster type for every state. It reveals that disasters like Flood and Storm are the most common

across many states, while some states show higher occurrences of specific disasters such as Hurricanes in Alabama and Floods in Arkansas. Rare events like Tsunami and Volcano appear very infrequently. This combined analysis helps in understanding state-wise disaster patterns and supports further visualization techniques like heatmaps and stacked bar charts.

7. State Comparison Chart.

This step focuses on visualizing the comparison of disaster occurrences across states using a bar chart. The code first selects the top 10 states with the highest number of disaster declarations using `head(10)` from the previously aggregated data. The `plot(kind='bar')` method from Pandas, along with Matplotlib, is used to create a bar chart representation.

The chart displays states on the x-axis and the number of disasters on the y-axis, making it easy to compare disaster frequency across the top states. The use of titles and axis labels improves clarity and readability. The output clearly highlights that states like Texas, Missouri, and Kentucky have the highest number of disaster declarations, indicating potential high-risk regions. This visualization helps in quickly identifying and comparing disaster-prone states.

8. Incident Type Distribution by State.

This step focuses on visualizing the distribution of different disaster types across states using a stacked bar chart. The code selects the first 10 states from the aggregated dataset and uses the `plot(kind='bar', stacked=True)` method to represent multiple disaster types within a single bar for each state.

In this chart, each bar represents a state, and the stacked segments within the bar represent different incident types, with their size indicating the number of occurrences. The use of stacking allows easy comparison of both total disasters and the contribution of each disaster type within a state. Titles, axis labels, and legends improve readability and interpretation.

The output shows that disaster types like Storm and Flood contribute significantly across most states, while certain states have higher proportions of specific disasters, such as Hurricanes in some regions. This visualization helps in understanding both overall disaster frequency and the composition of disaster types within each state.

9. Choropleth Map.

This step focuses on geospatial visualization using a choropleth map to represent disaster distribution across the United States. The code uses Plotly Express (`px.choropleth`) to create an interactive map where each state is colored based on the number of disaster declarations.

The `locations` parameter takes the state codes, while `color` represents the disaster count for each state. The `locationmode="USA-states"` ensures correct mapping, and `scope="usa"` limits the visualization to the United States. The map uses color intensity to indicate disaster frequency, where darker shades represent higher numbers of disasters.

The output is an interactive geographical map that clearly highlights disaster-prone states, such as Texas and other high-count regions. This visualization helps in easily identifying geographical hotspots and understanding spatial patterns, which are not easily visible in tabular data.

CONCLUSION.

The analysis reveals that natural disasters are unevenly distributed across the United States, with certain states experiencing significantly higher occurrences. States like Texas and Missouri emerge as major disaster-prone regions, while events such as Floods and Storms are the most common across multiple states. Geospatial visualizations, including bar charts and choropleth maps, effectively highlight these patterns and help identify high-risk areas. Overall, the study provides clear insights into disaster distribution, supporting better understanding and future planning.

